

Lecture 4:

Architectural Patterns

Software Architecture and Design Patterns

Lecture Outline

- What is a Pattern?
- Pattern Catalogue
 - Module patterns
 - Component and Connector Patterns
 - Allocation Patterns
- Summary

What is a Pattern?

An architectural pattern **establishes** a **relationship between**:

- *A **context***. A recurring, **common situation** in the world that **gives** rise to a **problem**.
- *A **problem***. The problem, appropriately **generalized**, that arises in the given context.
- *A **solution***. A successful **architectural resolution** to the **problem**, appropriately abstracted. The **solution** for a pattern is **determined** and described by:
 - **A set of element types** (for example, data repositories, processes, and objects)
 - **A set of interaction mechanisms** or **connectors** (for example, method calls, events, or message bus)
 - **A topological layout** of the components
 - A set of **semantic constraints** covering topology, element behavior, and interaction mechanisms

Pattern Catalogue

- Module Patterns
- Component-and-Connector Patterns
- Allocation Patterns